CS5275 The Algorithm Designer's Toolkit
S2 AY2025/26
Problem set 4

Submission is due at 23:59, April 9 (Thu).

Problem 1 (Learning an unknown parity function, 4 marks). Let $f : \{-1, 1\}^n \to \{-1, 1\}$ be a parity function. That is, there exists a subset $S \subseteq [n]$ such that

$$f(x) = \chi_S(x) = \prod_{i \in S} x_i.$$

You do not know f, but you may query f at inputs of your choice.

Design a *deterministic* algorithm that determines f in the fewest number of queries in the worst case. Prove that your algorithm is *correct* and *optimal*.

Problem 2 (BLR with four queries). Let $f : \mathbb{F}_2^n \to \{-1, 1\}$ be a Boolean function.

2.a (2 marks) Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{F}_2^n$ be chosen uniformly at random. Prove that

$$\mathbb{E}_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\big[f(\boldsymbol{x})f(\boldsymbol{y})f(\boldsymbol{z})f(\boldsymbol{x}+\boldsymbol{y}+\boldsymbol{z})\big] = \sum_{S \subseteq [n]} \widehat{f}(S)^4.$$

2.b (2 marks) Suppose f is ε-far from every parity function and its negation. That is, for every $S \subseteq [n]$,

$$\mathsf{Pr}_{\boldsymbol{x}}\big[f(\boldsymbol{x}) \neq \chi_S(\boldsymbol{x})\big] \geq \varepsilon \quad \text{and} \quad \mathsf{Pr}_{\boldsymbol{x}}\big[f(\boldsymbol{x}) \neq -\chi_S(\boldsymbol{x})\big] \geq \varepsilon,$$

where $\chi_S(x) = (-1)^{\sum_{i \in S} x_i}$.

Prove that

$$\mathsf{Pr}_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}}\big[f(\boldsymbol{x})f(\boldsymbol{y})f(\boldsymbol{z}) \neq f(\boldsymbol{x}+\boldsymbol{y}+\boldsymbol{z})\big] \in \Omega(\varepsilon).$$

Problem 3 (May's theorem, 4 marks). Let $f : \{-1, 1\}^n \to \{-1, 1\}$. If f is *monotone*, *odd*, and *symmetric* (see lecture 12 for definitions), prove that (i) n is odd, and (ii) f is the majority function Maj_n.

Problem 4 (Poincaré inequality and hypercube conductance, 4 marks). Prove that the conductance of the d-dimensional hypercube G is *exactly*

$$\Phi(G) = \frac{1}{d}.$$

Hint: You might want to use the Poincaré inequality (lecture 12):

$$\mathrm{Var}[f] \leq \mathrm{I}[f].$$